

Auteur: Noran Ben Said
Studierichting: Informatica
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$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{x+4}{x+2} \right)^{3x-1} \\ & \frac{x+2+2}{x+2} \Rightarrow \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x+2} \right)^{3x-1} \\ & = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{x+2} \right)^{\frac{x+2}{2}} \right]^{\frac{2}{x+2}(3x-1)} \\ & = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{x+2} \right)^{\frac{x+2}{2}} \right]^{\frac{6x-2}{x+2}} \\ & = e^6 \end{aligned}$$

References